

Self-attention is permutation equivariant

- $Y = \text{softmax}(XW_Q (XW_K)^\top / \sqrt{d}) XW_V$. For any permutation matrix P : **$f(PX) = P f(X)$**
- Every query sees the keys as a **set**: “the cat ate the mouse” and “the mouse ate the cat” give the same outputs, permuted
- Mean-pool the outputs $\rightarrow$ exactly **invariant** (T04 §5: error $\approx 10^{-16}$ in float64)
- Fix: inject the index i . Three places it can enter: the **input**, the **attention logits**, or the **queries and keys**
- The causal mask also breaks the symmetry: token i sees only i tokens ($\rightarrow$ NoPE)

Attention has no sense of order

self-attention without positions is permutation-equivariant

dog bites man"

dog

bites

man

output z

dog

bites

man

0.05

0.57

0.38

0.26

0.49

0.26

0.40

0.50

0.10

man bites dog"

man

bites

dog

output z

man

bites

dog

0.03

0.37

0.60

0.10

0.41

0.49

0.15

0.65

0.20

input = word embedding + 1.00 × position encoding

$\|z(\text{dog, first word}) - z(\text{dog, last word})\| = 1.581$

now the same word at a different position gets a different output: order matters

Course animation: PermutationEquivariance

Where position enters: three places

- **(a) Add to the input:** $x_i + p_i \rightarrow$ sinusoidal, learned absolute (Transformer, BERT, GPT-2, ViT)
- **(b) Bias the logits:** $e_{ij} + b(i - j) \rightarrow$ Shaw et al., T5 buckets, ALiBi
- **(c) Rotate q and k:** $e_{ij} = (R_i q_i)^\top (R_j k_j) \rightarrow$ RoPE (Llama, Qwen, Mistral, Gemma, DeepSeek)
- (a) mixes four terms in every layer; relative position $i - j$ never appears on its own:

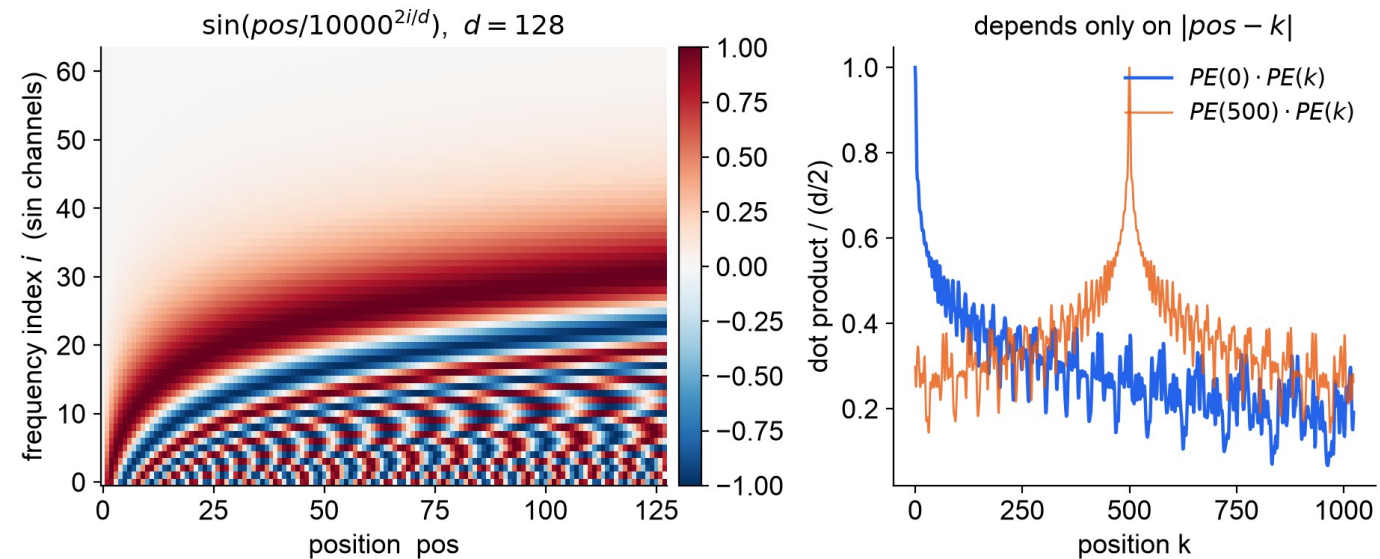
$$e_{ij} = \frac{(x_i + p_i)W_Q W_K^\top (x_j + p_j)^\top}{\sqrt{d}} = \underbrace{x_i W_Q W_K^\top x_j^\top}_{\text{content}} + \underbrace{x_i W_Q W_K^\top p_j^\top + p_i W_Q W_K^\top x_j^\top}_{\text{mixed}} + \underbrace{p_i W_Q W_K^\top p_j^\top}_{\text{position}}$$

(b) and (c) act inside attention in every layer and see only $i - j$. (c) is multiplicative: no content–position cross terms.

Sinusoidal position encoding (Vaswani et al., 2017)

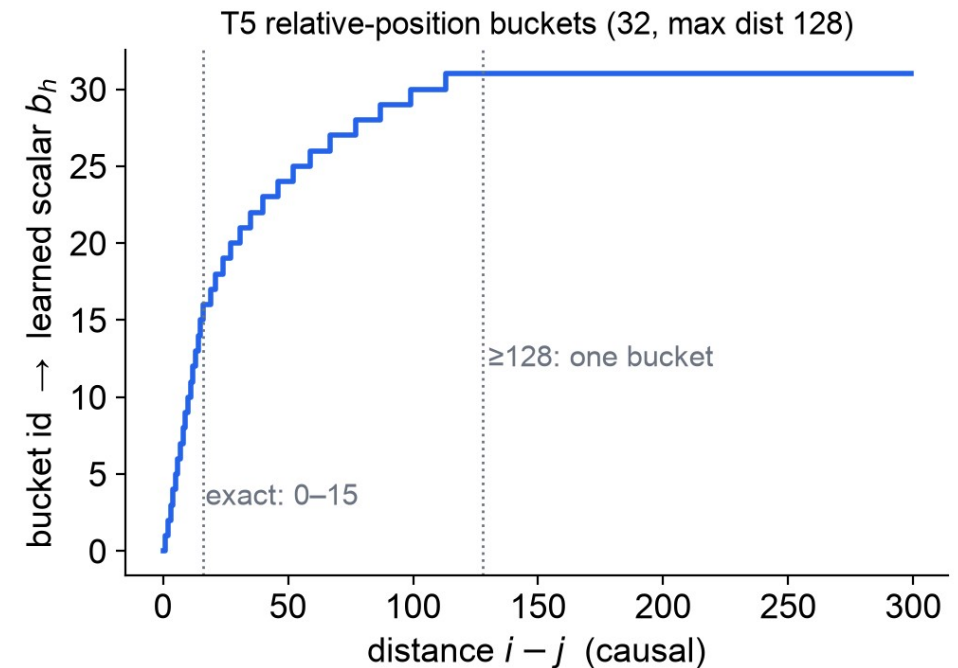
$$PE(pos, 2i) = \sin\left(pos/10000^{2i/d}\right), \quad PE(pos, 2i+1) = \cos\left(pos/10000^{2i/d}\right)$$

- $d/2$ frequencies $\omega_i = 10000^{-2i/d}$, geometric: wavelengths from 2π to $2\pi \cdot 10^4$ tokens
- Each (sin, cos) pair rotates by $k\omega_i$: $PE(pos + k) = M_k PE(pos)$, a fixed linear map independent of pos
- $PE(pos) \cdot PE(pos + k) = \sum_i \cos(k\omega_i)$ depends only on $|k|$ (right). After $W_Q W_K^\top$ the mixed terms bring absolute position back
- No parameters, defined for any pos; in practice it does not extrapolate past the training length (Press et al., 2022)



Learned absolute and relative positions

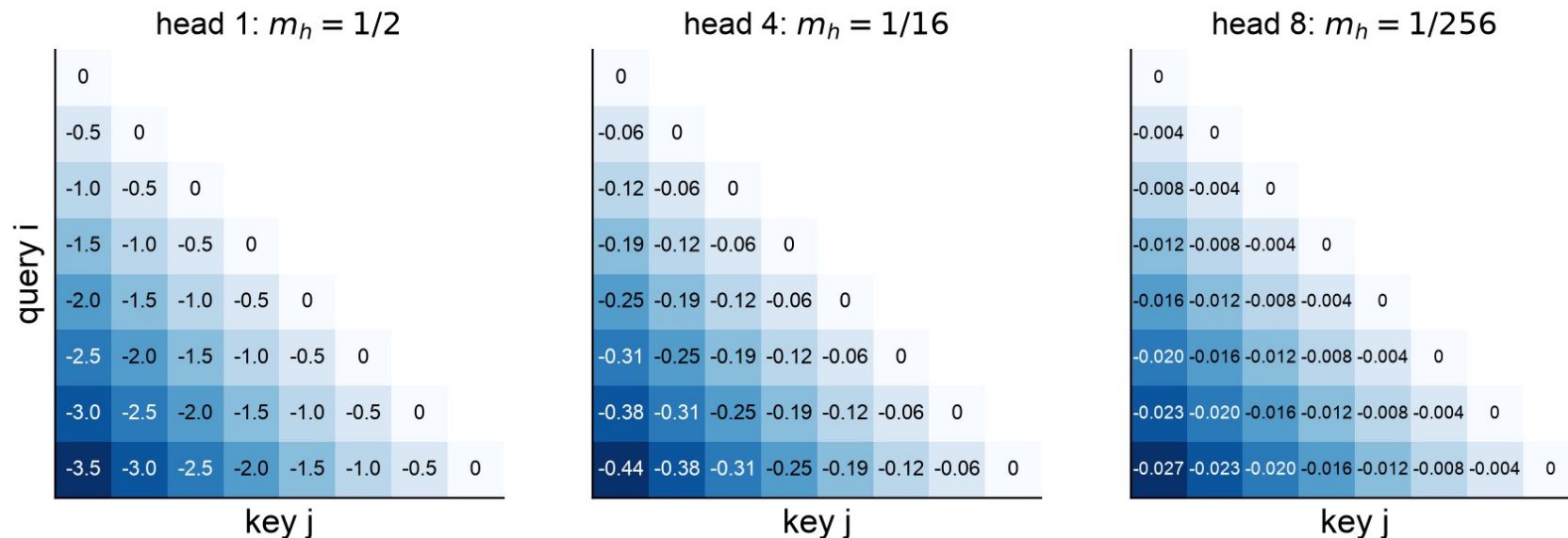
- **Learned absolute:** a table $P \in \mathbb{R}^{T_{\max} \times d}$, $x_i + P[i]$. BERT (512), GPT-2 (1024), ViT (one per patch; 2-D interpolated at a new resolution). Nothing is defined past $T_{\max}$
- **Shaw et al. (2018):** $e_{ij} = x_i W_Q (x_j W_K + a_{\text{clip}(j-i, k)})^\top / \sqrt{d}$ — one learned vector per offset clipped to $[-k, k]$, in every layer
- **T5 bias (Raffel et al., 2020):** $e_{ij} + b_h(\text{bucket}(i - j))$ — one scalar per head and bucket, shared across layers; 32 buckets: exact up to 16, log-spaced to 128, then one bucket
- Relative schemes are translation invariant, but they add an extra $T \times T$ term to every attention call



T5 bucket function (code from the T5 release), causal case

ALiBi: a linear recency bias (Press et al., 2022)

ALiBi bias $-m_h(i - j)$ added to causal logits ($H = 8, m_h = 2^{-8h/H}$)



- No position embedding at all: $e_{ij} = q_i \cdot k_j / \sqrt{d} - m_h(i - j)$, $j \leq i$; fixed slopes $m_h = 2^{-8h/H}$
- Zero parameters. Head 1 is very local, head 8 almost global
- “Train short, test long”: trained at 1024, perplexity stays flat at 2048+ (BLOOM, MPT)
- Cost: far tokens are always penalised, so retrieval over long distances is weaker than RoPE + extension

RoPE (1/5): rotate one pair of coordinates

- Goal: $\langle f(q, m), f(k, n) \rangle = g(q, k, m - n)$. The score sees content and offset only
- Sinusoidal fails (cross terms); Shaw/T5 are not an inner product of two vectors
- Take one 2-D pair and rotate q by $m\theta$, k by $n\theta$:

$$\begin{pmatrix} q'_{2j} \\ q'_{2j+1} \end{pmatrix} = \begin{pmatrix} \cos m\theta_j & -\sin m\theta_j \\ \sin m\theta_j & \cos m\theta_j \end{pmatrix} \begin{pmatrix} q_{2j} \\ q_{2j+1} \end{pmatrix}$$

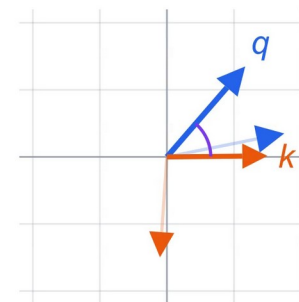
$$(R_m q)^\top (R_n k) = q^\top R_m^\top R_n k = q^\top R_{n-m} k$$

Rotations are orthogonal: $R_m^\top = R_{-m}$, $R_{-m} R_n = R_{n-m}$.
 $\|R_m q\| = \|q\|$: only the angle changes.

Rotary position embedding (RoPE)

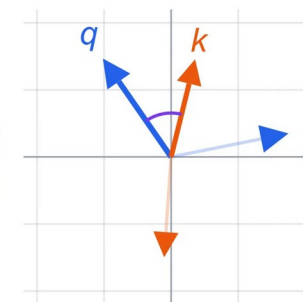
rotate q and k by an angle proportional to position

query at $m = 2$, key at $n = 5$



$q \cdot k = +1.772$

query at $m = 6$, key at $n = 9$



$q \cdot k = +1.772$

the angle between them is $(n - m) \cdot \theta$: the same offset gives the same score

Course animation: RoPERotation. Same offset, same score

RoPE (2/5): complex form, and why only $m - n$ survives

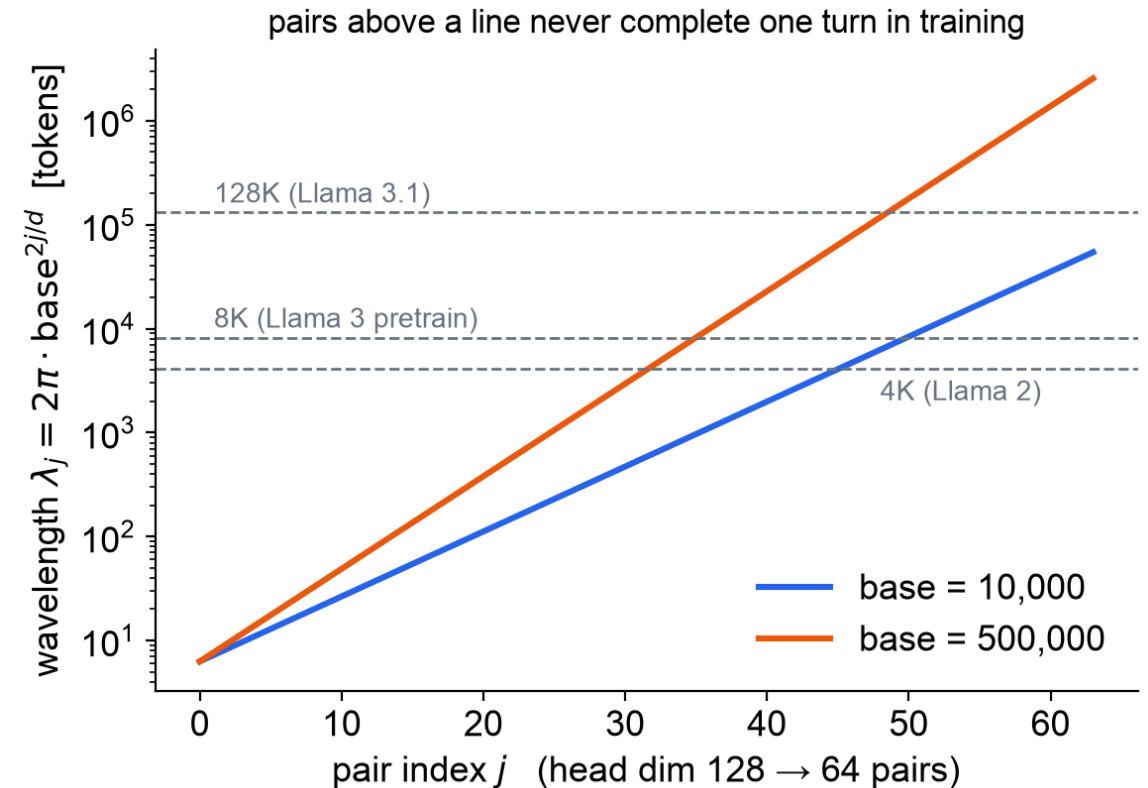
- Split the head dimension d into $d/2$ pairs and read each pair as a complex number: $\tilde{q}_j = q_{2j} + i q_{2j+1}$
- RoPE at position m : $\tilde{q}_j \mapsto \tilde{q}_j e^{im\theta_j}$. Multiplying by a unit complex number is a rotation by $m\theta_j$
- For 2-vectors a, b : $a^\top b = \text{Re}[\tilde{a} \tilde{b}^*]$ ($*$ = complex conjugate). So the score is

$$\langle R_m q, R_n k \rangle = \sum_{j=0}^{d/2-1} \text{Re} \left[\tilde{q}_j e^{im\theta_j} (\tilde{k}_j e^{in\theta_j})^* \right] = \sum_j \text{Re} \left[\tilde{q}_j \tilde{k}_j^* e^{i(m-n)\theta_j} \right]$$

- $e^{im\theta} (e^{in\theta})^* = e^{i(m-n)\theta}$: m and n appear only as $m - n$ ■
- Multiplicative, not additive: no content–position cross terms, absolute position never visible to the score
- As a matrix, $R_m = \text{diag}(R(m\theta_0), \dots, R(m\theta_{d/2-1}))$ is block-diagonal, so it is applied elementwise, never as a $d \times d$ matmul

RoPE (3/5): the frequency spectrum $\theta_j = \text{base}^{-2j/d}$

- $j = 0, \dots, d/2 - 1$; base = 10,000 in RoFormer. The same geometric ladder as sinusoidal PE
- Wavelength $\lambda_j = 2\pi / \theta_j$: from $2\pi \approx 6$ tokens ($j = 0$) to $\approx 54\text{K}$ tokens ($j = 63, d = 128$)
- **High-frequency pairs** (small j) wrap many times per context: fine, local offsets
- **Low-frequency pairs** (large j) turn less than once over the context: almost position-free, used as semantic channels (Barbero et al., 2025)
- Pairs with $\lambda_j > L_{\text{train}}$ never see a full turn in training: angles past L are out of distribution. Context extension fixes exactly these pairs
- Rule of thumb: $2\pi \cdot \text{base} \gtrsim \text{longest context}$. Llama 3: base 500,000

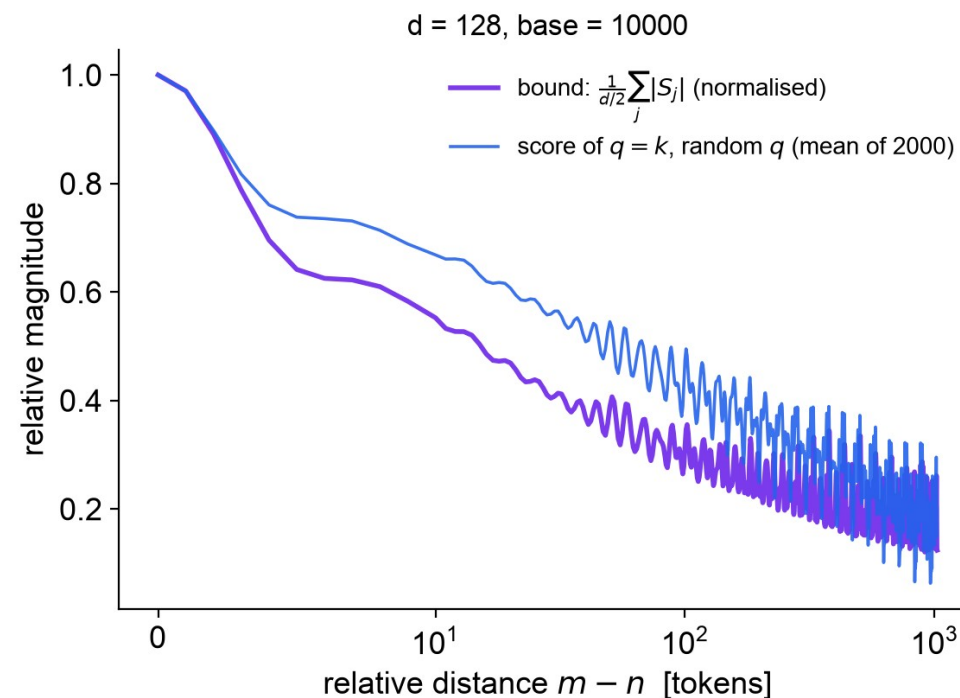


RoPE (4/5): long-range decay

- Write the score as $\sum_j h_j e^{i(m-n)\theta_j}$ with $h_j = \tilde{q}_j \tilde{k}_j^*$, and sum by parts (Abel):

$$\left| \sum_j h_j e^{i(m-n)\theta_j} \right| \leq \left(\max_j |h_{j+1} - h_j| \right) \sum_j |S_j|, \quad h_j = \tilde{q}_j \tilde{k}_j^*, \quad S_j = \sum_{l < j} e^{i(m-n)\theta_l}$$

- The average $(2/d) \sum_j |S_j|$ shrinks as $|m - n|$ grows (purple): with smooth h , far tokens get smaller logits
- Same-content pair $q = k$, random q (blue): the score drops from $\|q\|^2$ at offset 0 to $\approx 0.16\|q\|^2$ around 1K tokens (local average)
- A bound, not a law: trained heads can learn q, k that undo it (long-range heads lean on low-frequency pairs)



RoPE (5/5): implementation

- Rotate **q** and **k** after W_Q, W_K , per head, before $QK^\top$. **Not v**: position has already acted through the weights a_{ij}
- **cos/sin cache** of shape $(T_{\max}, d/2)$, computed once in fp32, indexed by position ids. No parameters, $O(T \cdot d)$ elementwise
- Two layouts: interleaved pairs $(2j, 2j+1)$ (RoFormer, Meta Llama) vs halves $(j, j + d/2)$, rotate-half (GPT-NeoX, HF). Same math up to a permutation; a mismatch silently breaks a checkpoint
- **KV cache stores rotated keys** $R_n k_n$: a key's position is fixed when written; a new query at step t is rotated by t only. Moving tokens inside the cache (eviction, StreamingLLM) needs re-rotation
- Position ids are just integers: packing, left padding and M-RoPE only change the ids

```
theta = base ** (-torch.arange(0, d, 2) / d) # (d/2,)
ang = torch.outer(torch.arange(T_max), theta)
cos, sin = ang.cos(), ang.sin() # cache once

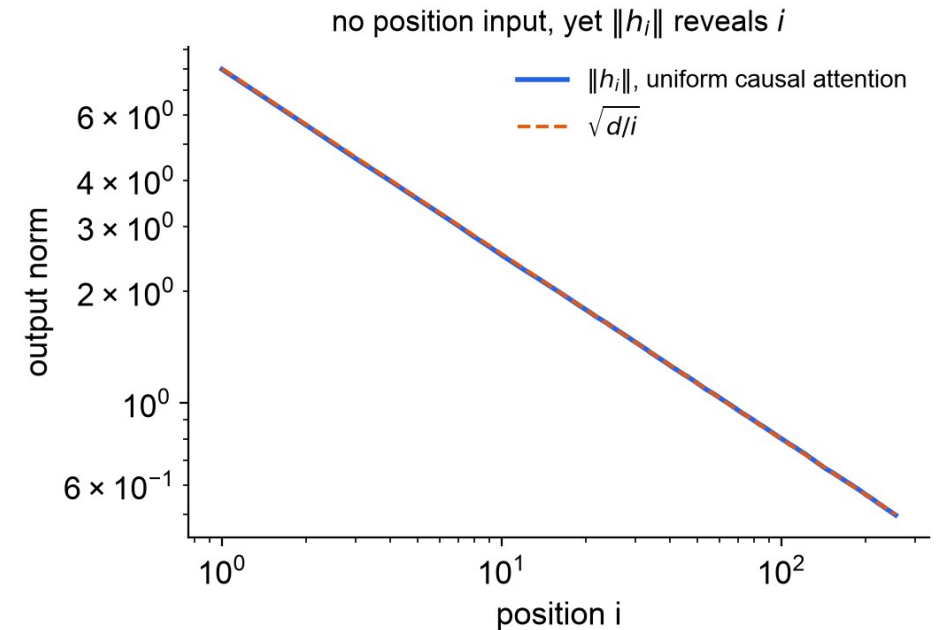
def rope(x, pos):
    # x: (B, H, T, d)
    x1, x2 = x[..., 0::2], x[..., 1::2]
    c, s = cos[pos], sin[pos]
    return torch.stack([x1*c - x2*s,
                        x1*s + x2*c], -1).flatten(-2)

q, k = rope(q, pos), rope(k, pos) # v untouched
cache.append(k, v)                # k already rotated
```

Same code as T04 §3 (rope_cache / apply_rope)

NoPE: no positional encoding at all

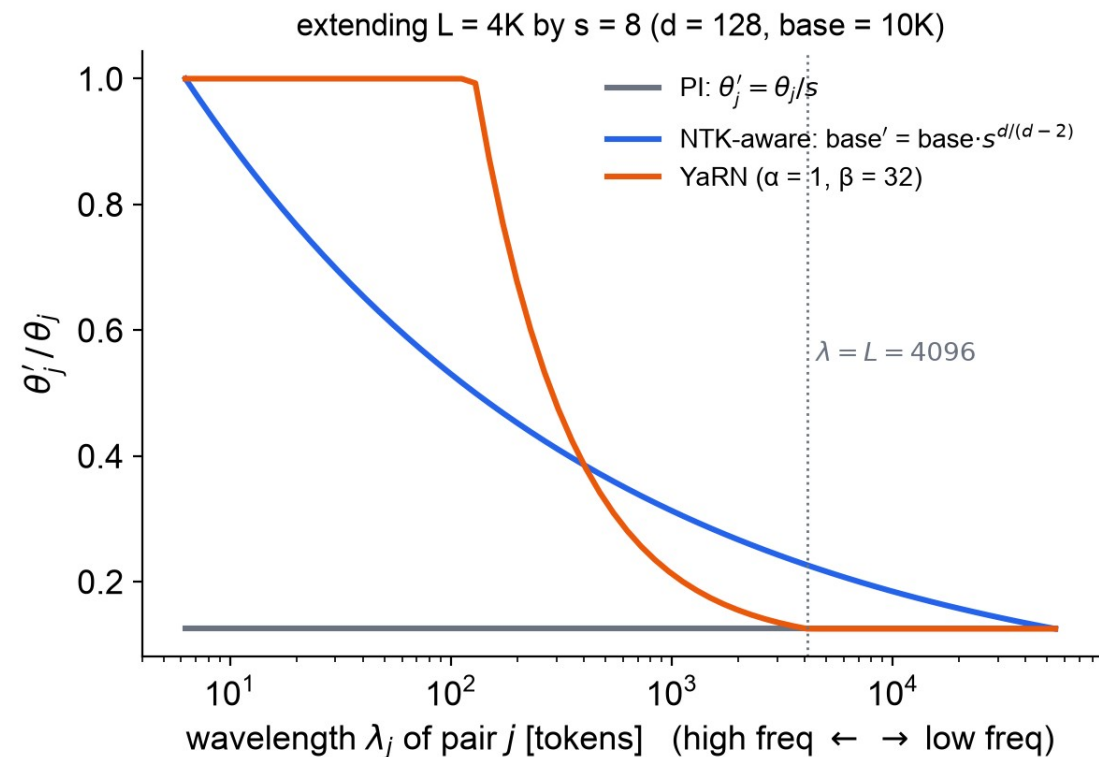
- Causal decoders trained with **no** positional encoding reach perplexity close to models with one (Haviv et al., 2022)
- Why: the causal mask breaks the permutation symmetry. Token i averages over i tokens, so statistics such as $\|h_i\| \approx \sqrt{(d / i)}$ encode i (right); later layers can read position from them
- Kazemnejad et al. (NeurIPS 2023): on small reasoning and arithmetic tasks NoPE length-generalises better than absolute PE, T5 bias, ALiBi and RoPE; it can represent both absolute and relative position
- Bidirectional encoders (BERT, ViT) cannot do this: no mask, nothing to count
- At scale NoPE alone lags; it is used for some of the layers (Cohere Command A, Llama 4)



Uniform causal attention over random tokens, $d = 64$, 500 sequences

Context extension: rescale the RoPE frequencies

- Train at L , run at $L' = s \cdot L$: the low-frequency pairs meet angles they never saw
- **Position Interpolation** (Chen et al., 2023): $m \mapsto m / s$, i.e. $\theta'_j = \theta_j / s$ for all j . Llama 2K $\rightarrow$ 32K with ~ 1 K fine-tuning steps; also squeezes the high-frequency pairs
- **NTK-aware** (bloc97, 2023): change only the base, $\text{base}' = \text{base} \cdot s^{d/(d-2)}$. High frequencies $\approx$ kept, the lowest pair scaled by s
- **YaRN** (Peng et al., ICLR 2024): per pair $r_j = L / \lambda_j$; $r_j < \alpha \rightarrow \text{PI}$, $r_j > \beta \rightarrow \text{keep}$, linear ramp between; plus logit temperature $\sqrt{(1/t)} = 0.1$ in $s + 1$. $10\times$ fewer tokens than earlier methods
- **Llama 3** (2024): base 500,000 from the start; 8K pre-training $\rightarrow$ 128K in six continued-pretraining stages (~ 800 B tokens)



Positions for images and video: 2-D RoPE and M-RoPE

- ViT: learned 1-D table (ViT), fixed 2-D sin-cos (MAE). **Axial 2-D RoPE**: half the pairs rotate by $x\theta_j$, half by $y\theta_j$, so the score depends on $(\Delta x, \Delta y)$ (Heo et al., ECCV 2024; DINOv3)
- **M-RoPE** (Qwen2-VL): split each head's pairs into three groups (t, h, w), each rotated by its own position id
 - text: $t = h = w = \text{index} \rightarrow$ exactly 1-D RoPE
 - image: t fixed, (h, w) = patch grid offset by the start id; video: t advances per frame
 - the next text token starts at max id + 1 (below)
- Qwen2.5-VL ties t to absolute time (seconds), not frame index. Open: which frequencies go to which axis (Qwen3-VL interleaves them)

	text			image (3 × 4 patches)												text	
t	0	1	2	3	3	3	3	3	3	3	3	3	3	3	3	7	8
h	0	1	2	3	3	3	3	4	4	4	4	5	5	5	5	7	8
w	0	1	2	3	4	5	6	3	4	5	6	3	4	5	6	7	8

Comparison

Scheme	Where it enters	Parameters	Extrapolates?	Used by
Sinusoidal	added to input x	0	poorly in practice	Transformer (2017)
Learned absolute	added to input x	$T_{\max} \cdot d$	no (undefined $> T_{\max}$)	BERT, GPT-2/3, OPT, ViT
Shaw relative	keys (and values), every layer	$(2k+1) \cdot d_h$ per layer	clipped offsets	early MT / music models
T5 bias	scalar bias on logits	$32 \cdot H$	partly (one far bucket)	T5, mT5
ALiBi	linear bias on logits	0	yes (perplexity)	BLOOM, MPT
RoPE	rotates q, k, every layer	0	no; yes with PI / NTK / YaRN	Llama, Qwen, Mistral, Gemma, DeepSeek
NoPE	nowhere (causal mask only)	0	best on small tasks	interleaved layers: Llama 4, Command A
2-D RoPE / M-RoPE	rotates q, k with (x, y) or (t, h, w) ids	0	via id design	DINOv3, Qwen2-VL / 2.5-VL

2025–26 updates and open questions

- **Position is now a per-layer choice.** Llama 4 “iRoPE”: RoPE in local (chunked) attention layers, interleaved global layers with no positional encoding, plus inference-time attention temperature scaling. Command A: RoPE + sliding window, every 4th layer full NoPE. Gemma 3: local layers base 10K, global layers base 1M
- **Partial RoPE:** rotate only part of each head (GPT-J, GPT-NeoX: 25%). Barbero et al. (ICLR 2025): models use the lowest frequencies as semantic channels; dropping them (p-RoPE) helps
- **MLA decoupled RoPE** (DeepSeek-V2/V3): keys are decoded from a compressed latent c^{KV} , and a rotation between W^{UK} and the query would block absorbing W^{UK} into W^Q . Fix: a separate 64-dim RoPE key shared by all heads, $q = [q^C ; R_m q^R]$, $k = [k^C ; R_n k^R]$
- **Open:** claimed vs usable context (needle tests pass, multi-hop at 128K fails); is the limit the encoding or the data? Do deep causal models need positions at all? How to split frequencies across t , h , w ?

Animations and tutorial

- Animations (animations/renderers/w04/):
 - PermutationEquivariance: permute the inputs, the outputs permute with them
 - WherePositionEnters: input (a), logits (b), q/k rotation (c)
 - SinusoidalClock: each (sin, cos) pair is a clock hand at its own speed
 - RoPERotation: same offset, same score (RoPE 1/5–2/5)
 - LengthExtrapolation: run past the training length; ALiBi, RoPE, PI / YaRN
- Tutorial T04_transformers.ipynb:
 - §3 “The modern block”: write rope_cache / apply_rope; check that $\langle R_m q, R_n k \rangle$ depends only on $n - m$, that the norm is preserved, and that it matches the complex form
 - §5 “Symmetry”: $f(x_\pi) = f(x)_\pi$ to 10^{-16} with no positions; RoPE breaks it, and so does the causal mask

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self-attention without positions is permutation-equivariant

"dog bites man"					"man bites dog"				
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bites	0.26	0.49	0.26		bites	0.10	0.41	0.49	
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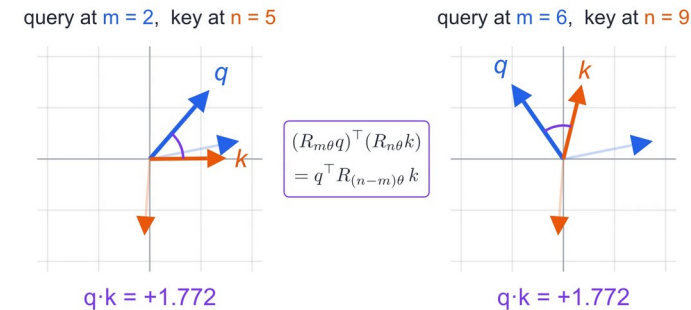
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the angle between them is $(n - m) \cdot \theta$: the same offset gives the same score